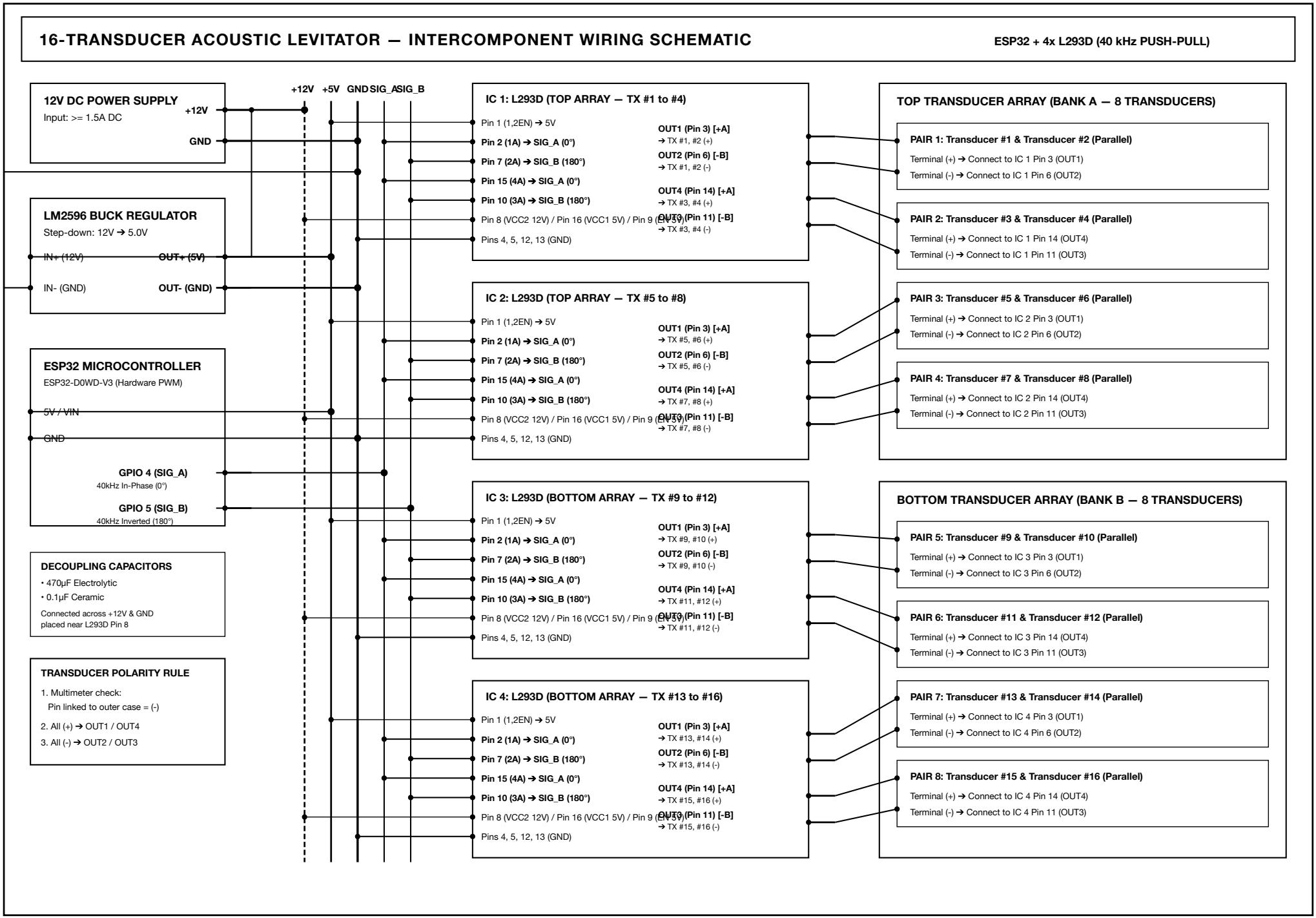

16-TRANSDUCER ACOUSTIC LEVITATOR — INTERCOMPONENT WIRING SCHEMATIC
ESP32 + 4x L293D (40 kHz PUSH-PULL)
12V DC POWER SUPPLY
Input: >= 1.5A DC
+12V
GND
LM2596 BUCK REGULATOR
Step-down: 12V → 5.0V
IN+ (12V)
OUT+ (5V)
IN- (GND)
OUT- (GND)
ESP32 MICROCONTROLLER
ESP32-D0WD-V3 (Hardware PWM)
5V / VIN
GND
GPIO 4 (SIG_A)
40kHz In-Phase (0°)
GPIO 5 (SIG_B)
40kHz Inverted (180°)
DECOUPLING CAPACITORS
• 470µF Electrolytic
• 0.1µF Ceramic
Connected across +12V & GND
placed near L293D Pin 8
TRANSDUCER POLARITY RULE
1. Multimeter check:
Pin linked to outer case = (-)
2. All (+) → OUT1 / OUT4
3. All (-) → OUT2 / OUT3
+12V
+5V
GND
SIG_A
SIG_B
IC 1: L293D (TOP ARRAY — TX #1 to #4)
Pin 1 (1,2EN) → 5V
Pin 2 (1A) → SIG_A (0°)
Pin 7 (2A) → SIG_B (180°)
Pin 15 (4A) → SIG_A (0°)
Pin 10 (3A) → SIG_B (180°)
Pin 8 (VCC2 12V) / Pin 16 (VCC1 5V) / Pin 9 (EN 5V)
Pins 4, 5, 12, 13 (GND)
OUT1 (Pin 3) [+A]
→ TX #1, #2 (+)
OUT2 (Pin 6) [-B]
→ TX #1, #2 (-)
OUT4 (Pin 14) [+A]
→ TX #3, #4 (+)
OUT3 (Pin 11) [-B]
→ TX #3, #4 (-)
IC 2: L293D (TOP ARRAY — TX #5 to #8)
Pin 1 (1,2EN) → 5V
Pin 2 (1A) → SIG_A (0°)
Pin 7 (2A) → SIG_B (180°)
Pin 15 (4A) → SIG_A (0°)
Pin 10 (3A) → SIG_B (180°)
Pin 8 (VCC2 12V) / Pin 16 (VCC1 5V) / Pin 9 (EN 5V)
Pins 4, 5, 12, 13 (GND)
OUT1 (Pin 3) [+A]
→ TX #5, #6 (+)
OUT2 (Pin 6) [-B]
→ TX #5, #6 (-)
OUT4 (Pin 14) [+A]
→ TX #7, #8 (+)
OUT3 (Pin 11) [-B]
→ TX #7, #8 (-)
IC 3: L293D (BOTTOM ARRAY — TX #9 to #12)
Pin 1 (1,2EN) → 5V
Pin 2 (1A) → SIG_A (0°)
Pin 7 (2A) → SIG_B (180°)
Pin 15 (4A) → SIG_A (0°)
Pin 10 (3A) → SIG_B (180°)
Pin 8 (VCC2 12V) / Pin 16 (VCC1 5V) / Pin 9 (EN 5V)
Pins 4, 5, 12, 13 (GND)
OUT1 (Pin 3) [+A]
→ TX #9, #10 (+)
OUT2 (Pin 6) [-B]
→ TX #9, #10 (-)
OUT4 (Pin 14) [+A]
→ TX #11, #12 (+)
OUT3 (Pin 11) [-B]
→ TX #11, #12 (-)
IC 4: L293D (BOTTOM ARRAY — TX #13 to #16)
Pin 1 (1,2EN) → 5V
Pin 2 (1A) → SIG_A (0°)
Pin 7 (2A) → SIG_B (180°)
Pin 15 (4A) → SIG_A (0°)
Pin 10 (3A) → SIG_B (180°)
Pin 8 (VCC2 12V) / Pin 16 (VCC1 5V) / Pin 9 (EN 5V)
Pins 4, 5, 12, 13 (GND)
OUT1 (Pin 3) [+A]
→ TX #13, #14 (+)
OUT2 (Pin 6) [-B]
→ TX #13, #14 (-)
OUT4 (Pin 14) [+A]
→ TX #15, #16 (+)
OUT3 (Pin 11) [-B]
→ TX #15, #16 (-)
TOP TRANSDUCER ARRAY (BANK A — 8 TRANSDUCERS)
PAIR 1: Transducer #1 & Transducer #2 (Parallel)
Terminal (+) → Connect to IC 1 Pin 3 (OUT1)
Terminal (-) → Connect to IC 1 Pin 6 (OUT2)
PAIR 2: Transducer #3 & Transducer #4 (Parallel)
Terminal (+) → Connect to IC 1 Pin 14 (OUT4)
Terminal (-) → Connect to IC 1 Pin 11 (OUT3)
PAIR 3: Transducer #5 & Transducer #6 (Parallel)
Terminal (+) → Connect to IC 2 Pin 3 (OUT1)
Terminal (-) → Connect to IC 2 Pin 6 (OUT2)
PAIR 4: Transducer #7 & Transducer #8 (Parallel)
Terminal (+) → Connect to IC 2 Pin 14 (OUT4)
Terminal (-) → Connect to IC 2 Pin 11 (OUT3)
BOTTOM TRANSDUCER ARRAY (BANK B — 8 TRANSDUCERS)
PAIR 5: Transducer #9 & Transducer #10 (Parallel)
Terminal (+) → Connect to IC 3 Pin 3 (OUT1)
Terminal (-) → Connect to IC 3 Pin 6 (OUT2)
PAIR 6: Transducer #11 & Transducer #12 (Parallel)
Terminal (+) → Connect to IC 3 Pin 14 (OUT4)
Terminal (-) → Connect to IC 3 Pin 11 (OUT3)
PAIR 7: Transducer #13 & Transducer #14 (Parallel)
Terminal (+) → Connect to IC 4 Pin 3 (OUT1)
Terminal (-) → Connect to IC 4 Pin 6 (OUT2)
PAIR 8: Transducer #15 & Transducer #16 (Parallel)
Terminal (+) → Connect to IC 4 Pin 14 (OUT4)
Terminal (-) → Connect to IC 4 Pin 11 (OUT3)